

KHARGA, Egypt

WMO: 624350

Lat: 25.4644N

Lon: 30.5425E

Elev: 239

StdP: 14.57

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.3	44.6	26.0	20.1	60.1	28.9	23.1	60.0	21.3	64.1	19.6	66.0	1.7	280	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.0	111.1	73.0	108.7	72.5	106.6	72.0	76.9	103.8	75.4	102.3	74.1	101.2	7.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	103.9	93.3	66.0	96.6	92.3	64.2	90.6	92.5	40.4	104.0	38.9	102.3	37.7	101.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.9	18.7	16.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	37.1	116.0	2.3	2.8	35.4	118.0	34.0	119.7	32.7	121.3	31.0	123.4	
			WB	34.2	78.8	2.9	2.2	32.1	80.4	30.4	81.6	28.7	82.9	26.6	84.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.1	59.4	63.2	70.0	79.0	86.7	91.0	91.8	92.2	88.8	81.6	70.8	62.0	(6)
	DBStd	13.05	4.83	5.92	6.61	6.73	6.07	4.51	3.57	3.49	4.39	5.68	5.88	4.79	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	437	189	94	21	1	0	0	0	0	0	0	12	120	(9)
	CDD50	10265	290	370	622	869	1138	1231	1297	1310	1163	978	625	372	(10)
	CDD65	5228	14	45	178	420	673	781	832	845	713	513	187	27	(11)
	CDH74	77589	204	593	2072	5393	10052	12747	13946	14064	10572	6021	1660	265	(12)
	CDH80	49344	40	210	940	3030	6425	8693	9690	9766	6683	3198	601	68	(13)
(14) Wind	WSAvg	6.8	5.2	6.0	7.0	7.6	7.6	8.2	6.4	6.7	8.0	7.7	6.2	5.4	(14)
(15) Precipitation	PrecAvg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(15)
	PrecMax	0.2	0.1	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.1	0.0	0.1	0.0	(16)
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.3	92.8	101.2	108.8	113.8	114.1	113.1	112.1	110.1	105.1	95.7	87.0	(19)
		MCWB	63.1	65.4	67.2	71.2	72.0	73.0	74.2	73.5	74.1	72.2	67.4	64.4	(20)
	2%	DB	79.6	86.1	95.0	103.8	109.0	110.3	109.7	108.8	106.2	100.3	90.8	80.9	(21)
		MCWB	59.9	62.0	64.3	69.2	70.9	72.3	73.5	73.3	72.7	70.6	66.3	61.9	(22)
	5%	DB	75.0	81.2	89.7	99.5	105.4	107.4	107.1	106.5	103.3	96.9	86.4	76.3	(23)
		MCWB	57.5	59.8	62.4	67.1	70.1	71.4	72.7	72.9	71.9	69.7	64.3	58.9	(24)
	10%	DB	71.3	76.4	84.6	94.9	101.6	104.4	104.9	104.5	100.4	93.5	82.8	72.9	(25)
		MCWB	56.0	57.7	60.7	64.9	69.0	70.5	72.0	72.4	71.2	68.3	62.7	57.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.3	67.1	68.8	74.0	75.9	77.6	78.7	78.9	77.4	74.8	70.3	67.2	(27)
		MCDB	80.5	89.0	96.4	102.6	106.0	107.3	107.4	103.5	102.9	99.0	90.2	83.6	(28)
	2%	WB	61.1	63.2	65.8	70.6	73.3	74.6	76.1	76.8	75.2	72.7	67.5	62.6	(29)
		MCDB	76.6	83.4	90.8	99.5	103.2	103.6	104.3	102.1	100.5	95.3	87.2	78.5	(30)
	5%	WB	58.6	60.5	63.5	68.3	71.4	72.9	74.6	75.2	73.6	70.9	65.4	60.0	(31)
		MCDB	73.5	78.8	87.4	96.3	101.5	102.6	102.6	101.3	99.1	93.1	84.0	74.1	(32)
	10%	WB	56.3	58.3	61.5	66.0	69.7	71.6	73.2	73.8	72.2	69.3	63.7	58.2	(33)
		MCDB	70.0	75.5	83.2	92.3	99.5	101.2	101.3	100.2	97.2	90.6	81.1	71.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.3	25.0	26.3	27.2	26.4	25.3	26.0	25.0	23.6	23.0	23.5	23.4	(35)
		MCDBR	29.6	31.7	34.4	34.0	31.8	29.3	29.7	27.3	26.5	26.1	27.0	27.9	(36)
		MCWBR	16.0	15.4	14.7	14.1	12.5	10.5	11.2	10.0	9.7	10.1	12.3	14.4	(37)
	5% WB	MCDBR	28.4	29.9	31.7	32.3	29.7	27.6	27.6	24.8	24.8	24.7	25.4	26.5	(38)
		MCWBR	16.2	15.7	14.6	14.4	12.6	10.8	11.4	10.0	10.3	10.6	12.5	14.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.386	0.423	0.473	0.567	0.585	0.513	0.505	0.501	0.491	0.485	0.410	0.389	(40)
	taud		2.057	1.936	1.811	1.628	1.611	1.780	1.819	1.837	1.844	1.841	2.007	2.050	(41)
	Ebn at Noon		269	268	261	240	235	251	253	254	253	245	259	262	(42)
	Edh at Noon		46	56	68	83	84	71	68	67	64	61	49	45	(43)
(44) All-Sky Solar Radiation	RadAvg		1442	1732	2073	2333	2476	2578	2505	2375	2144	1811	1524	1354	(44)
	RadStd		38	61	65	40	55	26	29	27	48	31	29	27	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (2 neighbors)	+0.54	N/A	N/A	N/A	-1.71	-3.06	N/A	N/A	N/A	N/A

Nomenclature: See separate page